Mahi Pasarkar

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https://github.com/mahi-pas | https://www.linkedin.com/in/mahi-pasarkar/ | https://rainyelephant.itch.io/

Languages: Python, Go, Typescript, Javascript, PostgreSQL, HTML5, PHP, Rust, C++, C#

Tools/Libraries: PyTorch, Numpy, Pandas, Matplotlib, NextJS, ReactJS, Django, Docker

Education

Rensselaer Polytechnic Institute Aug 2020 - May 2024

Bachelor of Science in Computer Science (AI and Data Concentration) and Games and Simulations (Dual Major) GPA: 3.7

Experience

Gradual Empowerment | Founder Sep 2026-Present

https://gradualempowerment.org/

- Running a 13-week Technical AI Safety course, GEAR, which covers a condensed version of the ARENA curriculum.
- Funded by a BlueDot Grant. First cohort had 250+ applicants and accepted 40.
- Hired and managed a TA and organized curriculum to emphasize reading papers and formulating extensions.

GEICO | Software Engineer II Jul 2024 - Present

- Managing a team of three and creating cloud observability system CapOpticon to monitor usage of GEICO's entire cloud. Tracking over 11,000 cloud resources to find anomalies in usage and suggest right-sizing. Found over 1,500 unused resources and identified 58,000 overprovisioned vCPU cores and 182,000 overprovisioned GiB of RAM.
- Built a daily Python ETL pipeline consolidating hardware and utilization data across distinct sources in GEICO's fleet of data centers, reconciling 775 physical servers into coherent data meant to understand our stock and empower purchasing and resource allocation decisions.
- Founded an innovation incubator giving employees time to work on unsolved problems, organized it for the Infrastructure Engineering department resulting in 36 new projects, 5 of which are now production systems.
- Mentoring two junior engineers from different departments, nurturing their engineering and career skills.

Research

Natural Language Processing for Structuring Disease Data | First Author Jan 2024 - Dec 2024

https://ntrs.nasa.gov/citations/20240015043

- First author, published by NASA and IEEE. Processed thousands of unstructured disease reports into machine readable format. Team Lead, designed system architecture in Python and divided work among coauthors to meet weekly OKRs.
- Engineered fault tolerant pipeline with extensible modules, logging, and debugging for future research use.

Projects & Miscellaneous

Project Duckweed | Sustainable Alternative Protein

- Identified Wolffia Globosa as an alternative protein candidate due to being 40% protein by weight and easy to grow.
- Wolffi Globosa is a hyperaccumulator of manganese and has a high iron content. To avoid exceeding the tolerable upper limit, I grew it in a closed hydroponics system. I learned about bioconcentration factors and designed my own fertilizer. By my calculation, the iron content would be too low to grow, but I ran the experiment to see if it could.
- Unfortunately, having below TUL of iron for 40g of protein is not possible, but I learned a lot about an unrelated field.

Jev as a CoT Monitor | https://www.lesswrong.com/posts/GHCNKiFzDYThTELFp/jev-as-a-cot-monitor-6x-faster-and-500x-cheaper

- Compared Jev with Sonnet 5 and GPT 5.6 Luna at identifying harm from 2,200 thought traces from ReasoningShield.
- Jev had comparable performance, but was 6x faster and 500x cheaper, showing promise as a control tool at scale.

Working Theories NYC | Founder | https://workingtheories.nyc

- Organizing social games in NYC meant to congregate high openness thinkers. Examples include *Hivemind*, which restricted language to only refer to a collective, or *Judge a Book By Its Cover*, which tested first-impression accuracy.
- Led large groups of over 30 people, garnering over 250 RSVPs for some events.